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Generalized voltage estimation of TCSC-compensated transmission lines for adaptive distance protection

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ABSTRACT

The variable performance of the thyristor-controlled series compensator (TCSC) system represents a vital challenge for conventional distance relays. Most existing algorithms are based on the measured data at two ends of the TCSC-compensated lines. However, some of single-end algorithms need to modify the distance relay for the series capacitor (SC) only. This paper introduces a generalized algorithm to estimate the voltage drop across the TCSC. The distance relay uses the estimated TCSC voltage drop for terminal voltage compensation. The second objective of the proposed algorithm is to estimate the conduction angle of the thyristor-controlled reactor (TCR) branch under different modes of operation. This algorithm is effective for different TCSC modes such as the capacitive or the inductive modes, in addition to the method of firing angle of synchronization current or voltage synchronization modes. The main benefit of the proposed algorithm is presenting a parameter-less distance scheme for TCSC-compensated transmission line. A 400-kV overhead transmission line test system is simulated in MATLAB/SIMULINK package® and the numerical modeling is implemented in the MATLAB® script in order to evaluate the proposed algorithm effectiveness. The proposed technique is tested under several stress conditions that may face the conventional distance relay. The results illustrate the effective performance of the proposed algorithm.

1. Introduction

The Flexible AC transmission systems (FACTS) have a main impact on the controllability modes of the transmission lines (TLs) [1]. The FACTS devices improve power system reliability and sustainability [1,2]. Moreover, they can increase the TL capacity and reduce its power losses. Series compensators (SCs) are widely used with FACTS devices that are used with long TLs. The SCs are used to increase power transfer capability, improve steady-state and transient stability limits, enhance voltage stability, and reduce power losses [3–5]. The thyristor-controlled series compensator (TCSC) is considered one of the SC devices, which is installed in series with TLs to change their parameters to improve the transmission capability or damp the sub-synchronous oscillations [6,7].

The metal oxide varistors (MOVs) are used to protect the series capacitor in the TCSC device to clamp the voltage drop across the capacitor during fault periods [3]. The operation of MOV causes several protection problems [8]. Distance protection is subjected to over-reach problems in the presence of SC due to the increase of fault current through MOV during fault periods [9].

Distance relays are subjected to some limitations due to high fault resistance and system condition changes. The effect of the fault resistance in relay tripping is discussed in [9]. It is noted that the active and reactive power flow is the main factor that affects distance relay trips. It is well known that the TCSC can change the parameters of the transmission system; so it can effectively control the power flow in the transmission system. Moreover, the dynamic nature of TCSC adds a serious challenge to the distance relay system. Therefore, it is very essential to install adaptive distance protection, which considers the operation of MOV, level of compensation and system condition changes [3].

Despite the benefits of installing TCSC in the power system, the installation position, controlling parameters and its structure might affect the apparent impedance detected by the protective device [10].

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The conventional distance characteristic is greatly subjected to maloperation in the presence of TCSC. The maloperation is due to under-reaching or over-reaching from the fault point. Consequently, the distance relay with conventional characteristics might not properly achieve the protective mission, where TCSC devices are installed [11–13]. Furthermore, the TCSC has a negative impact on steady-state and transient voltage and fault current components during fault periods [14]. This problem is initiated due to the variable capacitance or inductive impedances in series with the protected TL. It affects the impedance estimated by the distance relay, which leads to under-reach or over-reach problems [15–17]. The estimation of voltage across the TCSC during various system conditions is very vital to adapt the operation of the distance relay.

To overcome this problem introduced by TCSC, the data exchanging among the relays at the line terminals with a communication channel is developed [11–13,18–23]. In [18], the fault location is determined by a special algorithm based on two-terminal fault-locator developed for compensated TL with TCSC. The algorithm is applied by developing time domain model of both TL and TCSC. The fault location technique contains two stages considering the fault direction, which may be upstream or downstream TCSC device. Through the first cycle after fault moment, this technique determines the fault location by the identification of TCSC equivalent reactance. In [19], the fault location can be estimated depending on an algorithm which uses synchronized two-end fault-locator. By this algorithm, the estimation of fault location is achieved by taking into account that the fault direction is downstream or upstream the installed compensator. In [20], for series and shunt-compensated TLs, a pilot relaying scheme that depends on the incremental reactive power coefficients is presented. In [21], a fault location estimation scheme is presented based on superimposed currents. This technique enables to overcome reverse over-reaching of distance relays in TCSC-compensated TLs. Thus, this relay algorithm uses support vector machine and random forest classifiers to extract essential properties from the signals of current. This solution is costly and not reliable as well as the processing time is increased, with the need for communication channels [22]. On the other hand, a mathematical modification is required in the distance relay by compensating the relay terminal voltage. This is accomplished by estimating the series capacitor (SC) voltage under faults beyond the SC location, which modifies the apparent impedance [11,12,23]. A voting system based on the angles of the conventional and modified impedances together is used to discriminate if the fault is downstream or upstream the SC [11]. This solution is preferable, but it exhibits a shortage in its validity for SC type only. Because, [11] considers the analysis of the uncontrolled series device consists of MOV in parallel with a fixed capacitor. In the case of controlled series compensation, the analysis should be developed to consider the controlled reactor involved in this compensator.

In [15], a dynamic impedance mathematic model for TCSC-full current limiter (TCSC-FCL) is presented. The analysis includes two modes of operation: the series compensation mode and the current-limiting mode. A time-varying dynamic impedance model of TCSC-FCL is derived and analyzed. A strategy of distance protection setting considering the dynamic offset impedance circle is suggested to effectively reduce the impact of TCSC-FCL dynamic characteristics on the operation of distance protection. A bypass mode in the first cycle with TCSC-FCL is considered, which enhances the system performance during the fault transient periods. The bypass control mode is used to avoid discontinuity through the transition from maximum capacitive to maximum inductive modes. The main drawback of this research is the need for communication between the distance relay and the TCSC-FCL device. Also, a delay time is needed to avoid the bypass mode transient periods. In addition, the voltage synchronous control mode is considered while the current synchronous control mode is not considered during the analysis of TCSC-FCL model derives.

This paper presents a general algorithm to estimate the voltage drop across the TCSC from the relaying point. The measured voltage of the distance protection is modified with the TCSC estimated voltage drop to be suitable for the upstream faults from TCSC location. This algorithm is valid whether the TCSC is in the capacitive or the inductive modes and whether it works in current or voltage synchronization modes. This is accomplished by estimating the conduction angle of the TCR branch, which is the second contribution in this paper. The introduced mathematical approach is applied and embedded for the distance relay without the requirement to other devices. Simulation results using MATLAB/SIMULINK® package ensure the proposed algorithm efficacy under different TCSC controlling modes and fault conditions. The proposed algorithm is investigated under various noise levels and sampling frequencies to illustrate its effectiveness and robustness. The developed technology is integrated with the adaptive distance relay and tested under different power system stress conditions.

2. The TCSC main components and principle of operation

Fig. 1 illustrates the construction of the TCSC. It consists of a capacitor (C) connected in parallel with thyristor-controlled reactor (TCR), which includes reactor (L) in series with antiparallel thyristors. Metal-oxide varistor (MOV) is installed in parallel with the capacitor to protect it against high voltages [24,25]. In Fig. 1 (a), CB can bypass the device to control its use. To limit the transient current when the capacitor is bypassed, a current-limiter reactor L_d is installed in series with the CB. Ultra-high-speed switch (UHSC) is used to limit the operating losses due to the thyristor switches if the reactor is used for a long time. The installation time of the TCSC is less than the time required for the construction of other FACTS devices [6]. TCSC can control the power flow through the TL and raise the transmission capability instead of the construction of an additional TL. In this paper, the analysis of the system is based on the basic model shown in Fig. 1(b), which consists of TCR, fixed capacitor and MOV [26,27].

The control of the equivalent reactance of the TCSC is achieved by controlling the firing angle α, which may be synchronized to the zero crossing of the line current (Synchronized with current) [28] or synchronized to the peak value of the reversal voltage on the capacitor terminals (Synchronized with voltage) [29]. Fig. 2 illustrates the block diagram of the synchronization process referring to line current (a) (Syn-C) or the reversal voltage on capacitor (b) (Syn-V). The main two modes of the TCSC are the capacitive mode and inductive mode. Between these two modes, the resonance mode is avoided to be applied due to the instability within this mode [26].

3. System description

The simple single-line diagram of the compensated TL by TCSC is illustrated in Fig. 3. The TCSC is assumed to be installed in the middle of the TL connected between two substations A and B. Appendix A provides the different parameters of the studied system as well as the TCSC and MOV parameters [11,12].

The steady-state analysis of the TCSC enables to determine the required compensation characteristic. Fig. 4 illustrates the line current

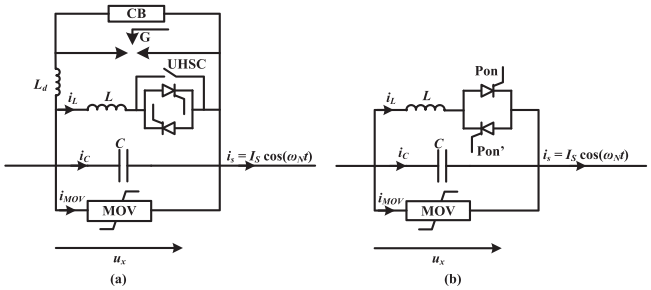


Fig. 1. TCSC models (a) practical (b) basic.

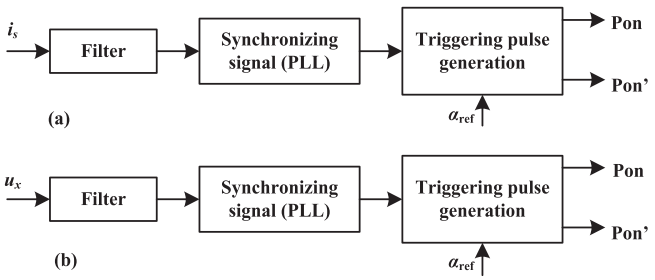


Fig. 2. TCSC synchronizing methodologies (a) Syn-C (b) Syn-V.

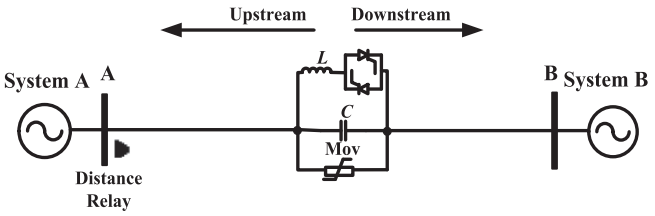


Fig. 3. Single line diagram of studied system compensated by TCSC.

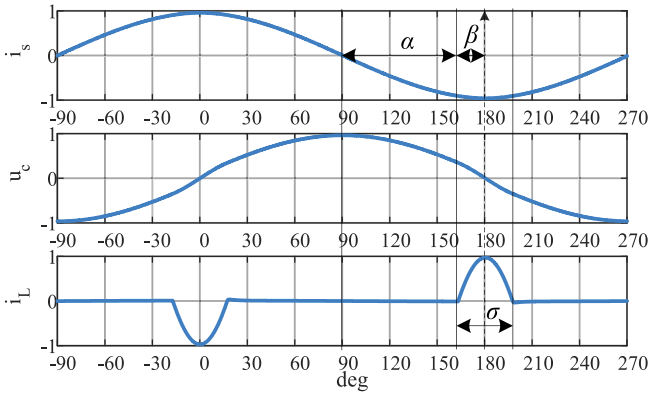


Fig. 4. Generic waveforms for TCSC operated in capacitive mode.

waveforms i_s, the capacitor voltage u_C and the inductor current i_L. The capacitor bank *(C)* and the inductor *(L)* form an *LC* circuit with a resonance frequency of ω_0:

$$\omega_0 = \frac{1}{\sqrt{LC}} \tag{1}$$

For appropriate operation of the TCSC, it is preferred to set $\omega_0 > \omega_N$, where ω_N is the nominal system frequency. Therefore, a design parameter $\lambda = \omega_0/\omega_N$, which is the ratio between ω_0 and ω_N, is set to fall in the range between 2 and 4 [26]. As shown in Fig. 4 in steady state condition, the thyristor firing angle (α) is used as a control variable. It is common to refer to α at the last instant when the maximum voltage is applied to the thyristor (Synchronized with voltage). In steady state operation, the phase shift between the line current and fundamental voltage of the capacitor is 90°. Therefore, α can refer to the zero-crossing of the line current (Synchronized with current). Practically, it is common to use the control angle β (shown in Fig. 4) to analyse the operation of the TCSC. The relations between α, β and the conduction angle σ are as the following:

$$\alpha = \pi - \beta \tag{2}$$

$$\sigma = 2\beta \tag{3}$$

The equivalent reactance (x_{eq}) of the TCSC can be determined, neglecting losses, according to Eq. (4) [29]:

$$x_{eq} = \frac{1}{j\omega_N C}\left\{1 + \frac{2}{\pi}\frac{\lambda^2}{\lambda^2-1}\left[\frac{2\cos^2\beta}{\lambda^2-1}(\lambda\tan\lambda\beta - \tan\beta) - \beta - \frac{\sin 2\beta}{2}\right]\right\} \tag{4}$$

According to the mentioned parameters of the TL in Appendix A, the TCSC is designed to compensate the TL from 50% to 100% of the line reactance. As shown in Fig. 5, the TCSC can operate in both capacitive and inductive modes, although the inductive mode is rarely used practically. According to the TCSC characteristic, the resonance happens around 47°. Therefore, it is preferred to prohibit the operation within the range 40° to 57° to provide adequate stability to the system. The capacitive mode is achieved with α from 57° to 90°. The capacitive reactance is a minimum at 90° with x_{eq} equals 50 Ω, which presents 50% compensation of the line inductive reactance. On the other hand, at firing angle of 57°, the equivalent capacitive reactance equals 100 Ω, which presents 100% of the total line inductive reactance. The inductive mode provides series inductive reactance ranging from 9.7 to 100 Ω corresponding to firing angles from 0 to 40°, respectively.

4. The proposed adaptive algorithm for TCSC voltage estimation

4.1. General description of the proposed methodology

The determination of the fault location of the series compensated TL requires the identification of the voltage across the TCSC compensation device [12]. The accurate estimation of the voltage across the TCSC from the relay point requires proper and accurate mathematical modeling of the device and TL. In this work, various operation circumstances are taken into accounts such as the mode of operation (capacitive or inductive) and the method of firing angle synchronization (Synchronized with current or Synchronized with voltage).

As shown in Fig. 6, the measured voltage and current at the relay point with the reference firing angle enable to determine the voltage of the TCSC. Nevertheless, the synchronization mode and the fault occurrence affect the conduction angle of the TCR and, therefore, affect the voltage estimation. Consequently, the modeling of the system should consider the mutual effect between the TCSC estimated voltage and TCSC actual conduction angle. The estimated voltage helps to compensate the voltage at the relay point and, then, helps to determine the impedance estimated from the relay point. If the estimated impedance is located within a certain protection zone, the trip order can be applied.

4.2. Numerical model of the TCSC voltage drop estimation

The objective of the TCSC voltage estimation is to eliminate the effect

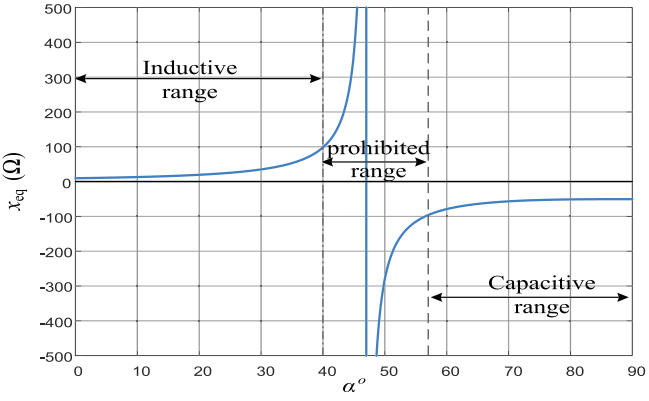


Fig. 5. Characteristics of the installed TCSC.

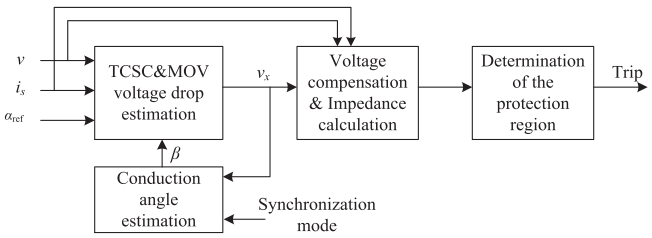


Fig. 6. Block diagram of the general methodology.

of TCSC compensator on the conventional distance relay. This is accomplished by deriving a mathematical formulation of the TCSC compensator voltage drop. The new estimated voltage at the relay point is the difference between the measured voltage and the estimated TCSC voltage drop [11,12].

As illustrated in Fig. 1(b), the TCSC consists of three parallel components: a fixed capacitor C, a reactor L in series with antiparallel thyristors (TCR) and MOVs. The MOV voltage-current characteristic is modeled by Eq. (5), which represents a non-linear resistor [12]. Eq. (6) shows the current passing through the series capacitor that depends on the first derivative of the voltage across the TCSC. Also, the TCR current is represented by integrating the voltage across TCSC as given by Eq. (7). This is occurred only if the antiparallel thyristors are triggered related to a compensation value.

$$i_{MOV} = P\left(\frac{v_x}{v_{ref}}\right)^q \tag{5}$$

$$i_C = c\frac{dv_x}{dt} \tag{6}$$

$$i_L = \frac{1}{L}\int_{-\beta}^{\beta} v_x dt \tag{7}$$

where i_{MOV}, i_C and i_L are MOVs, series capacitor and TCR currents, respectively; v_x is the MOVs voltage; v_{ref} is the reference voltage of the MOVs; q and P are MOVs parameters. In normal operation described in Fig. 4, TCSC has two durations per half cycle. The first duration is considered when the line current passes through the series capacitor and the MOVs only, which is represented by differential Eq. (8) [11,12]. On the other hand, the second one is considered when the line current passes through the three elements. This duration interval is $-\beta < \omega_N t < \beta$ that is described by the differential-integral Eq. (9).

$$c\frac{dv_x}{dt} + P\left(\frac{v_x}{v_{ref}}\right)^q - i_s = 0, \beta < \omega_N t < \pi - \beta \tag{8}$$

$$c\frac{dv_x}{dt} + P\left(\frac{v_x}{v_{ref}}\right)^q + \frac{1}{L}\int_{-\beta}^{\beta} v_x dt - i_s = 0, \ -\beta < \omega_N t < \beta \tag{9}$$

where: i_s is the TL current.

The differential and differential-integral Eqs. (8) and (9), respectively, should be reformed from the continuous-time formulas into their algebraic discrete-time formulas. The second order Gear differentiation rule is applied to transfer the differential equation to discrete equation [12]. Therefore, to estimate the voltage drop on the TCSC (v_x), the following equations should be applied:

$$i_s(t) \rightarrow i_s(n), v_x(t) \rightarrow v_x(n) \tag{10}$$

$$\frac{dv_x(t)}{dt} \rightarrow \frac{1}{2h}(3v_x(\mathrm{n}) - 4v_x(\mathrm{n}-1) + v_x(\mathrm{n}-2)) \tag{11}$$

while the definite integral estimation using a trapezoidal rule is proposed and applied in this study as follows:

$$\int_{-\beta}^{\beta} \mathrm{v_x(t)dt} \rightarrow \frac{\mathrm{h}}{2}\begin{pmatrix} v_x(\mathrm{n}-\mathrm{N}_\beta) + v_x(\mathrm{n}) \\ +2(v_x(\mathrm{n}-\mathrm{N}_\beta - 1) \\ +v_x(\mathrm{n}-\mathrm{N}_\beta - 2) + \cdots \\ +v_x(\mathrm{n}-2) + v_x(\mathrm{n}-1) \end{pmatrix} \tag{12}$$

where $h = T_s$; T_s is the sampling period; n is the discrete time index; and N_β is the number of samples within the period $[-\beta, \beta]$. From Eqs. (10), (11) and (12), the differential-integral Eq. (9) can be rewritten in the algebraic discrete-time form, which is considered in interval $-\beta < \omega_N t < \beta$, as the following proposed equation:

$$\frac{C}{2h}\begin{pmatrix} 3v_x(n) \\ -4v_x(n-1) \\ +v_x(n-2) \end{pmatrix} + P\left(\frac{v_x(n)}{v_{ref}}\right)^q + \frac{h}{2L}\left[+2\begin{pmatrix} v_x(\mathrm{n}) + v_x(\mathrm{n}-N_\beta) \\ v_x(\mathrm{n}-N_\beta+1) \\ +v_x(\mathrm{n}-N_\beta+2) \\ +\cdots + v_x(\mathrm{n}-2) \\ +v_x(\mathrm{n}-1) \end{pmatrix}\right] - i_s(n) = 0 \tag{13}$$

Eq. (13) should be reordered as in Eq. (14) to construct the form given by Eq. (15) as follows:

$$P\left(\frac{v_x(n)}{v_{ref}}\right)^q + \left(\frac{3C}{2h} + \frac{h}{2L}\right)v_x(n) - \begin{pmatrix} i_s(n) + \frac{C}{2h}\begin{pmatrix} 4v_x(n-1) \\ -v_x(n-2) \end{pmatrix} \\ -\frac{h}{2L}\begin{bmatrix} v_x(\mathrm{n}-N_\beta) \\ +2(v_x(\mathrm{n}-N_\beta+1) \\ +v_x(\mathrm{n}-N_\beta+2) + \cdots \\ +v_x(\mathrm{n}-2) + v_x(\mathrm{n}-1) \end{bmatrix} \end{pmatrix} = 0 \tag{14}$$

$$F(y(n)) = A_q(y(n))^q + A_1 y(n) - A_0 = 0 \tag{15}$$

where $y(n)$ is the voltage of TCSC&MOVs p.u., which equals $\frac{v_x(n)}{v_{ref}}$. To estimate the voltage across the TCSC&MOVs numeric Newton method is applied. The discrete-time solution of Eq. (15) is described by the following equations [27]:

$$y(n) = y(\mathrm{n}-1) - \frac{A_q.y(n-1)^q + A_1.y(n-1) - A_0}{q.A_q.y(n-1)^{q-1} + A_1} \tag{16}$$

where, in interval $\beta < \omega_N t < \pi - \beta$, the three parameters of this equation are defined as:

$$A_q = P \tag{17}$$

$$A_1 = \left(\frac{3C}{2h} + \frac{h}{2L}\right)v_{ref} \tag{18}$$

$$A_0 = \begin{pmatrix} i_s(n) + \\ \frac{3C}{2h}\begin{pmatrix} 4v_x(n-1) \\ -v_x(n-2) \end{pmatrix} \end{pmatrix} - \frac{h}{2L}\begin{bmatrix} v_x(\mathrm{n}-N_\beta) \\ +2\sum_{k=1}^{N_\beta - 1} v_x(\mathrm{n}-k) \end{bmatrix} \tag{19}$$

Similarly, from Eqs. (10) and (11), the differential Eq. (8) can be rewritten in the algebraic discrete-time form which is considered in the interval $\beta < \omega_N t < \pi - \beta$. From Eq. (14) and by neglecting the part that is related to the TCR current, Eq. (16) can be used after the modification of the three parameters as:

$$A_q = P \tag{20}$$

$$A_1 = \frac{3C}{2h}v_{ref} \tag{21}$$

$$A_0 = i_s(n) + \frac{C}{2h}(4v_x(n-1) - v_x(n-2)) \tag{22}$$

4.3. Algorithm of the proposed technique of TCSC voltage estimation

Fig. 7 illustrates the flowchart of the proposed algorithm to be followed by the distance protection of the TCSC-compensated TL. Continuously, the algorithm reads the voltage and current at the relay location. According to the TCSC mode of operation, type of synchronization and the firing angle, the estimated voltage across the TCSC can be determined and, then, the current through each element can be calculated. The estimation of TCR current enables to determine the actual conduction angle needed in the voltage estimation. The estimated voltage of the TCSC compensates the voltage at the relay terminal. Therefore, according to the calculated impedance, it can be identified whether it is a faulty or healthy case. If the system is in the faulty case, according to [11], the location can be determined. The voltage estimation process is continued until the end of the delay time to provide the trip command.

Using Discrete Fourier Transform (DFT), the values of positive, negative and zero sequence components of voltages and current can be determined from the instantaneous measured voltages and currents. According to the detected sequence component, the system state (healthy or faulty) and the fault type can be recognized [11]. According to any compensated voltage ($V_x(n)$), the adaptive fault impedance ($ZR(n)$) is calculated as:

$$ZR(n) = \frac{V_{sel}(n) - V_{xsel}(n)}{I_{sel.(n)}} \tag{23}$$

where, $V_{sel}(n)$ represents the selected voltage; $V_{xsel}(n)$ is the compensated selected voltage and $I_{sel.(n)}$ is the selected current at sample number (n) consistent with the fault type. For example, for line-to-ground faults, the $I_{sel.(n)}$ is modified by $I_a + k{\cdot}I_0$, where I_a is the faulty phase current, I_0 is the zero-sequence current and k is a compensation factor that equals $\frac{Z_0 - Z_1}{Z_1}$. As in [11], the identification of the faulty side enables to select the correct impedance and to give a trip signal after the appropriate operating time. The identification of the TCSC voltage is directly obtained from the DFT signal according to the specified measurement window without any iterations in the calculation process.

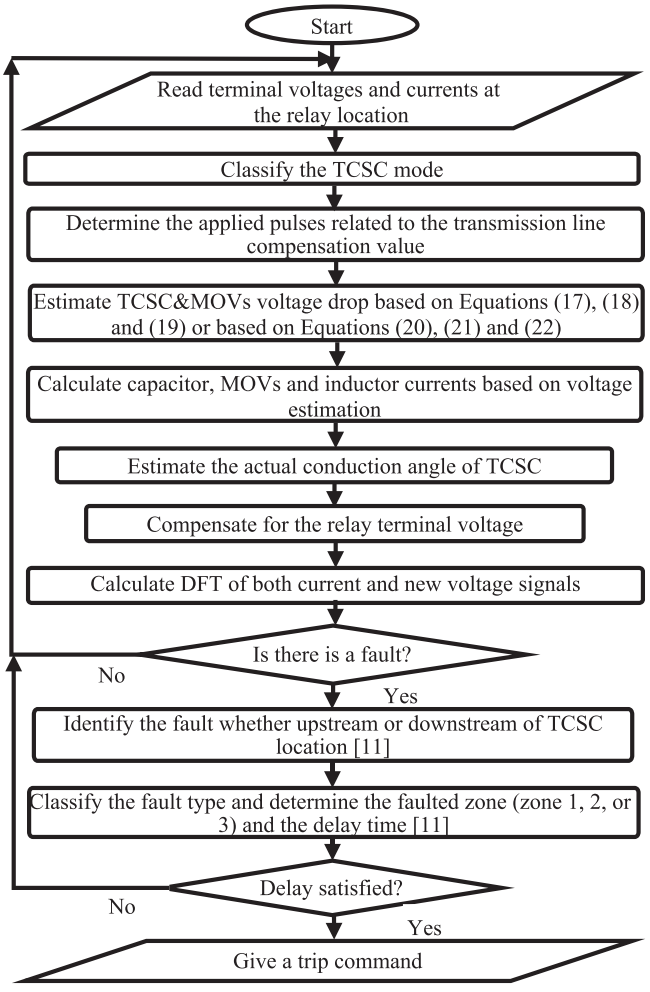


Fig. 7. Flowchart of the proposed distance relay.

The main contribution of this work is to develop a general algorithm to estimate the voltage drop across the TCSC from the relaying point. Then, this method is incorporated in distance protection for the compensated transmission lines. However, the used mathematical core of the adaptive distance relay fault impedance (Eq. (23)) is deduced by neglecting the fault resistance. Therefore, an error exists in the obtained impedance under faults associated with fault resistance. To overcome this situation, the fault resistance and the current of the remote end must be taken into consideration during the derivation of the mathematical core of the contemporary distance relay. This increases the complexity of the protection system. In this paper, the fault location is obtained via dividing the imaginary part of the calculated impedance per line reactance per unit length, to reduce the fault location error.

5. Results and discussion

Based on the algorithm and mathematical model developed in the previous section, MATLAB/SIMULINK package® simulation model is constructed to implement the relay algorithm. A sampling frequency of 20 kHz is used during the simulation of the model at different modes of operation [11]. The conventional distance relay has a good performance for faults upstream the installed TCSC without a need for compensation of the voltage drop across the series compensator. However, it is essential to determine the voltage across the TCSC in the case of downstream faults. The developed algorithm depends on the compensation of the measured voltage by the voltage drop across the TCSC and MOVs as in Eq. (16). The results discuss different modes of operation of TCSC under solid 3-phase fault happened at 180 km downstream the TCSC and with different firing angles as illustrated in the following subsections. Therefore, the algorithm is tested for capacitive and inductive modes in case of firing angles synchronized with current and voltage. In addition, it is essential to make sure that the algorithm operates efficiently even for cases close to the resonance mode. In this study, the fault inception instant is assumed to be at 3 s.

5.1. Capacitive mode synchronized with current (C-Syn-C)

The capacitive mode of TCSC is a mode where the compensator produces capacitive effect according to the equivalent reactance as shown in Fig. 5. Synchronization with current means that the firing angle is applied referring to the the zero crossing of the line current.

The three-phase current signals at relay point, at bus A, are shown in Fig. 8 under solid three-phase fault occurred while the TCSC operates at $\alpha = 70^\circ$ (between 57° and 90°). Figs. 9 and 10 illustrate that both the measured and estimated MOVs, series capacitor and TCR currents are very close to each other. The applied pulses and the estimated conduction angle, which is calculated from the inductor current, are shown in Fig. 11. The conduction angle has been associated with the duration of inductor current. It can be noticed that during the steady state operation, the applied pulses coincide with the estimated pulses. Nevertheless, after the fault instant, the inductor current appears only in one half cycle, and disappears in other periods. This is due to the synchronization of pulses with current, while during the fault period, the phase shift between the TCSC current and the voltage does not equal 90° as in the healthy system.

The changing of the phase shift depends on the TL parameters, the

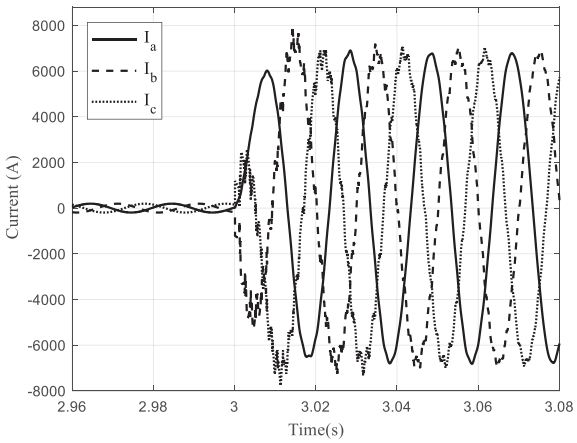


Fig. 8. The waveform of the currents at Bus A under solid 3-phase fault at 180 km and TCSC with 70° firing angle and C-Syn-C.

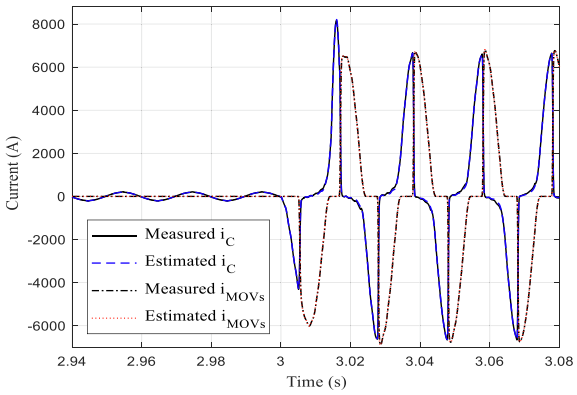


Fig. 9. Capacitor and MOVs currents waveforms under solid 3-phase fault at 180 km and TCSC with 70° firing angle and C-Sync-C.

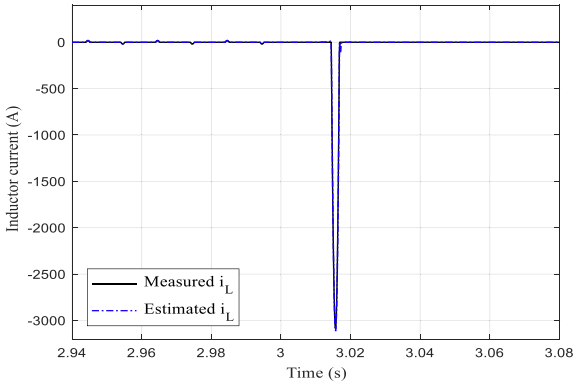


Fig. 10. Inductor current waveform under solid 3-phase fault at 180 km and TCSC in phase A with 70° firing angle and C-Syn-C.

TCSC parameters and the mode of operation. The estimated voltage drop across the TCSC matches the actual voltage drop as shown in Fig. 12. The measured voltage through the TCSC is obtained from the SIMULINK® simulation model, while the estimated value is determined by the developed numerical algorithm by MATLAB® script. A statistical evaluation is performed to compare between the measured and estimated TCSC voltage to illustrate the validity of the proposed algorithm. The percentages of maximum, minimum, mean, and standard deviation errors (where, $error = \frac{Measured\ \ value - Estimated\ \ value}{Measured\ \ value}$) of the estimated voltage drop at 70° firing angle are 1.5054%, 8.83 × 10^{-4}%, 0.8897%, and

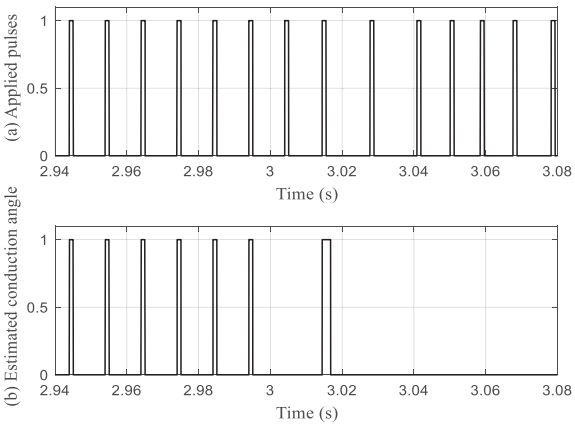


Fig. 11. Applied pulses and actual conduction angles in phase A under three-phase fault at 180 km with 70° firing angle and C-Syn-C.

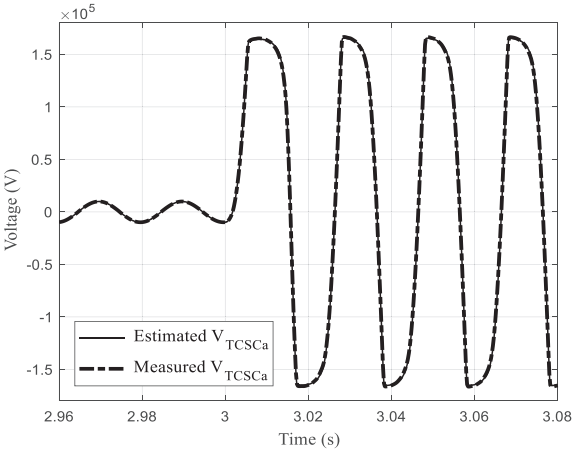


Fig. 12. Measured and estimated voltage drop across TCSC&MOV in phase A with 70° firing angle and C-Syn-C.

0.4374%, respectively.

As mentioned above, capacitive mode is achieved with α from 57° to 90°. Also, the estimated voltage drop across the TCSC is very close to the measured voltage drop for 90° firing angles. The estimation of the voltage in this case follows the developed Eqs. (20), (21) and (22). The maximum, minimum, mean, and standard deviation errors of the estimated voltage drop at 90° firing angle are 2.2817, 4.42 × 10^{-4}, 0.8786, and 0.8571, respectively. In addition, these values at, 80° firing angle are 1.5054, 8.84 × 10^{-4}, 0.8897, and 0.4374, respectively.

By using the estimated TCSC voltage drop, the measured relay voltage can be compensated as in [11]. Therefore, the fault location can be determined as shown in Fig. 13 by dividing the estimated impedance by unit length impedance. The obtained distances are inside zone 1 setting, which equals 80% of the total TL length. The measured lengths are 180.1023, 180.102 and 180.0678 for TSCR firing angles of 70°, 80° and 90°, respectively. Moreover, the measured impedance angle is around 85.2544° for different firing angles indicating their locations in the first zone setting.

5.2. Capacitive mode synchronized with voltage (C-Syn-V)

In this mode, the pulses are synchronized with the voltage on the TCSC. As shown in Fig. 2(b), the voltage across the device is filtered using phase locked loop (PLL) technique and, then, the pulses are applied referring to the voltage peak value.

Fig. 14 illustrates that the measured and estimated inductor currents

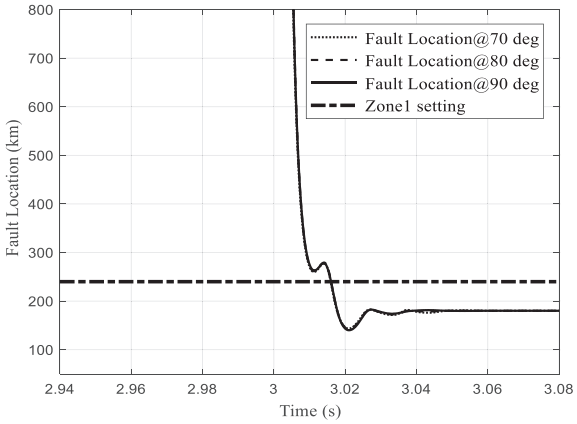


Fig. 13. Estimated fault distance and the setting of zone 1 under solid 3-phase fault at 180 km and different firing angle.

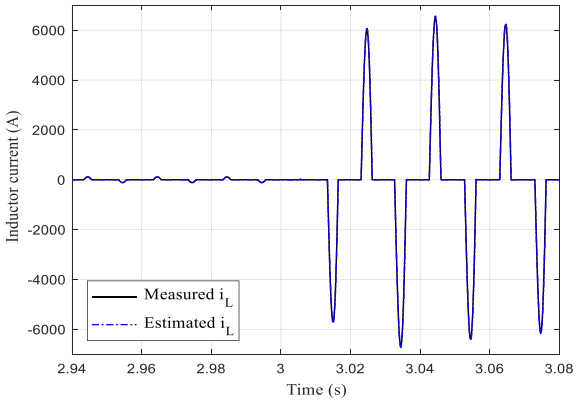


Fig. 14. Inductor current waveform under solid 3-phase fault at 180 km and TCSC with 70° firing angle and C-Syn-V.

are approximately identical. On the contrary of the Syn-C, during the fault, the inductor current exists except for the first half cycle due to the delay of the synchronizing signal filtration. The actual conduction angle tracks the inductor current as in Fig. 15. In the steady-state period before fault, both the applied pulses and the actual conduction angle are the same regarding to the inductor current duration. Before and after fault instant, the proposed developed algorithm succeeded to estimate the voltage drop across the TCSC with high accuracy as shown in Fig. 16. The percentages of maximum, minimum, mean, and standard deviation

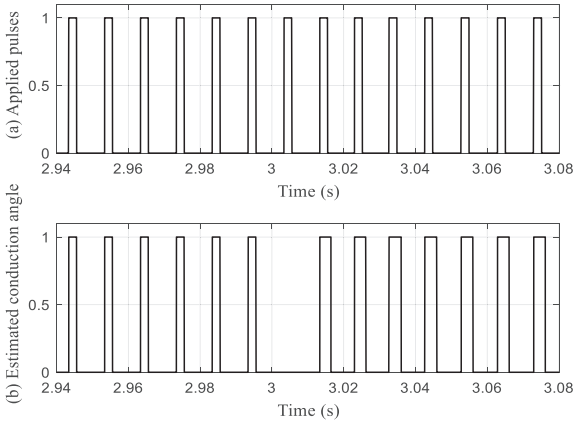


Fig. 15. Applied pulse and actual conduction angle in phase A under solid 3-phase fault at 180 km and C-Syn-V.

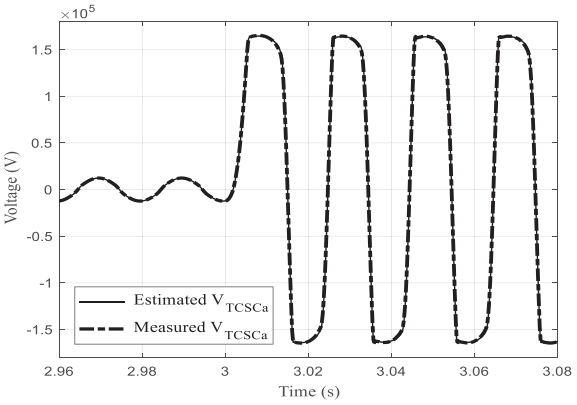


Fig. 16. Measured and estimated voltage drop across TCSC&MOV in phase A with 70° firing angle and C-Syn-V.

errors of the estimated voltage drop at 70° firing angle (C-Syn-V) are 1.5070%, 9.11 × 10^{-4}%, 0.7721% and 0.4549%, respectively.

5.3. Inductive mode synchronized with current (I-Syn-C)

The inductive mode of TCSC is a mode where the compensator produces inductive effect according to the equivalent reactance as shown in Fig. 5. The firing angle ranges from 0 to 40°.

The three-phase current signals at relay point, at bus A, are shown in Fig. 17 for solid three-phase faults, while the TCSC operates at $\alpha = 40°$. It can be noticed that the TCSC limits the line currents in the inductive mode. Figs. 18 and 19 illustrate that the currents of measured and estimated MOVs, series capacitor and TCR are very close to each other. The current limiting effect in the inductive mode delays the operation of MOVs with two complete cycles. The applied pulses and the estimated conduction angle, which is calculated from the inductor current, are shown in Fig. 20. The conduction angle has been associated with the duration of inductor current. It can be noticed that, before the fault instant, the applied pulses coincide with the estimated pulses. Nevertheless, after the fault instant, the inductor current is synchronized with the applied pulses after one and half period.

Fig. 21 illustrates both the estimated and measured values of the TCSC at 40° over the whole periods. The percentages of maximum, minimum, mean, and standard deviation errors of the estimated voltage drop at 40° firing angle are 2.6%, 5.76 × 10^{-4}%, 2.0884% and 0.7439%, respectively.

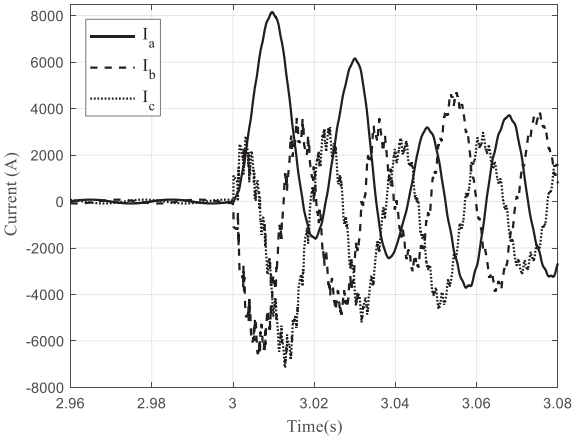


Fig. 17. Three phase current waveforms at Bus A under solid 3-phase fault at 180 km and TCSC with 40° firing angle and I-Syn-C.

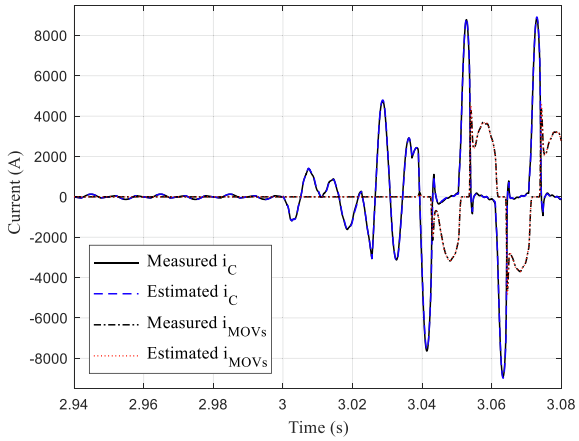


Fig. 18. Capacitor and MOVs current waveforms under solid 3-phase fault at 180 km and TCSC with 40° firing angle and I-Syn-C.

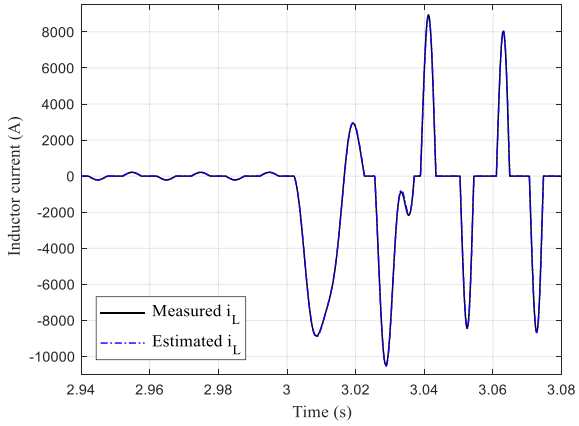


Fig. 19. Inductor current waveform under solid 3-phase fault at 180 km and TCSC with 40° firing angle and I-Syn-C.

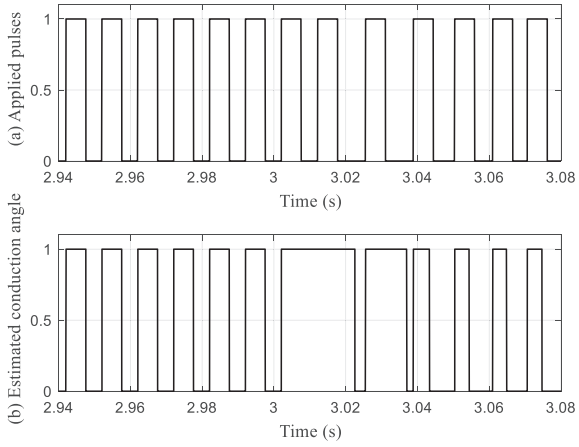


Fig. 20. Applied pulses and actual conduction angle in phase A under solid 3-phase fault at 180 km and I-Syn-C.

5.4. Inductive mode synchronized with voltage (I-Syn-V)

In this section, the developed algorithm is used to estimate the voltage across the TCSC in I-Syn-V. Fig. 22 shows the estimated and measured currents through the capacitor and MOV at a firing angle of 40°. Fig. 23 illustrates the TCR current, where after the fault instant, the

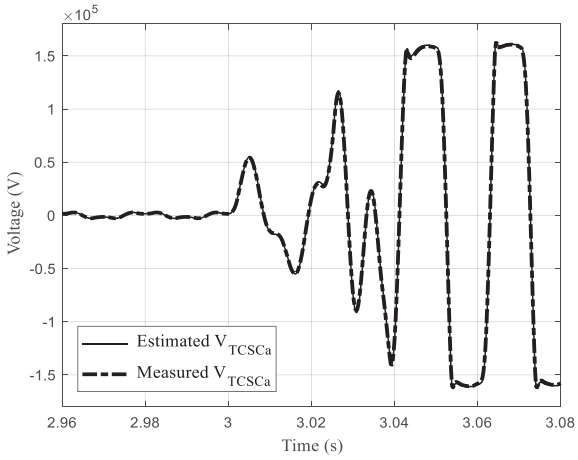


Fig. 21. Measured and estimated voltage drop though TCSC&MOV with 40° firing angle in phase A and I-Syn-C.

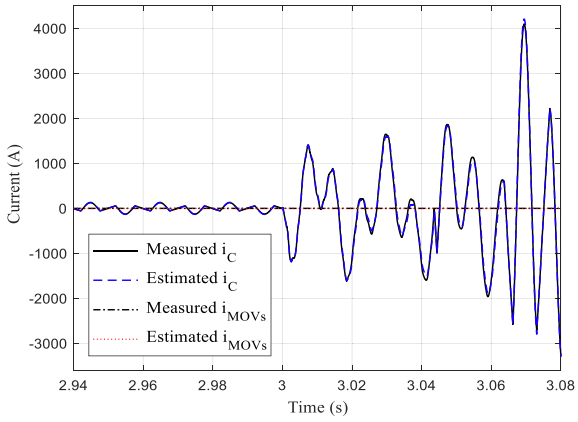


Fig. 22. Capacitor and MOVs currents waveforms under three-phase fault at 180 km and TCSC with 40° firing angle and I-Syn-V.

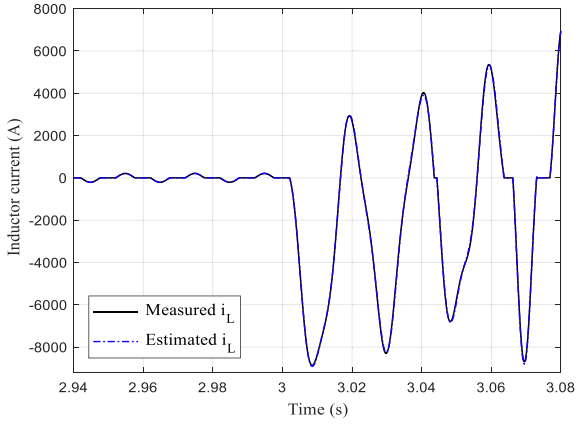


Fig. 23. Inductor current waveform under three-phase fault at 180 km and TCSC with 40° firing angle and I-Syn-V.

inductor current is approximately continuous for three periods and, then, the conduction angle coincides with the applied pulses as shown in Fig. 24. Fig. 25 illustrates the measured and estimated TCSC voltages with the following statistical values: The percentages of maximum, minimum, mean, and standard deviation errors of the estimated voltage drop at 40° firing angle are 1.9037%, 0.0035%, 1.3420%, and 0.4097%, respectively. It can be noticed that the MOVs current approximately

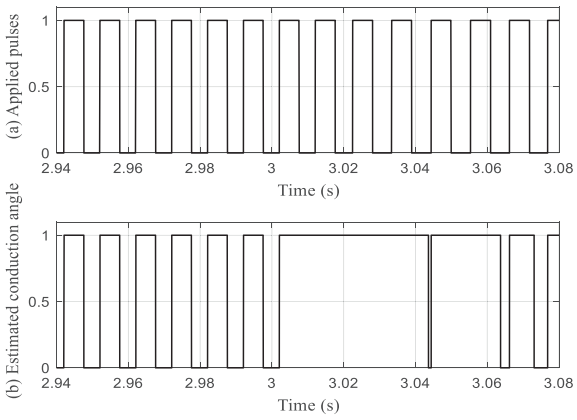


Fig. 24. Applied pulses and actual conduction angle in phase A under three-phase fault at 180 km and TCSC with 40° firing angle and I-Syn-V.

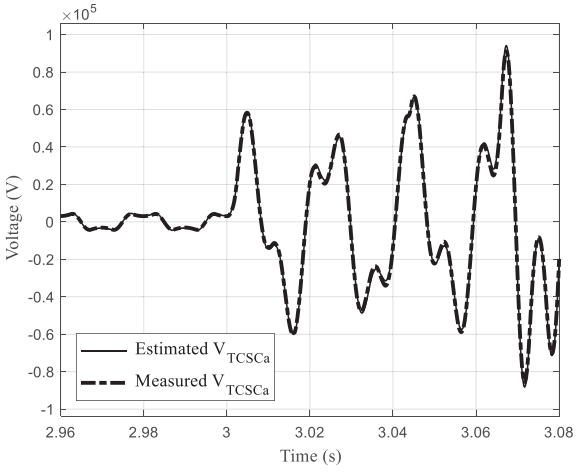


Fig. 25. Measured and estimated voltage drops though TCSC&MOV with 40° firing angle in phase A and I-Syn-V.

equals zero, because of the inductive mode and the synchronization with voltage. Therefore, the TCSC voltage during fault is less than 150 kV as shown in Fig. 25.

5.5. Resonance mode

Although it is very rare to use the TCSC close to the resonance case (prohibited range) shown in Fig. 5, this mode of operation may be used as a fault current limiter [30]. The studied TCSC operates at resonance at a firing angle of 47°. The developed algorithm is used to estimate the voltage across the TCSC around the resonance case (±2°). At 49°, the TCSC operates in capacitive mode and it may operate in Syn-C and Syn-V where Fig. 26 illustrates the operation at C-Syn-C mode. At 45°, the TCSC operates in the inductive mode and it may operate in Syn-C. Fig. 27 illustrates the estimated and measured voltages in I-Syn-V mode. The voltage synchronization mode prevents the appearance of high voltage across the TCSC as in Fig. 27. Therefore, the MOV is not activated in these cases.

Table 1 shows the statistical evaluation of the developed algorithm by calculating the maximum, minimum, mean errors and standard deviation between the Measured and estimated voltage across the TCSC. The algorithm is evaluated at different modes of operation and several firing angles. The results prove the proper accuracy of the developed algorithm to estimate the voltage across the TCSC under different modes of operation. From this table, the maximum error does not exceed 2.6% for all studied cases in Table 1, which is considered as an acceptable

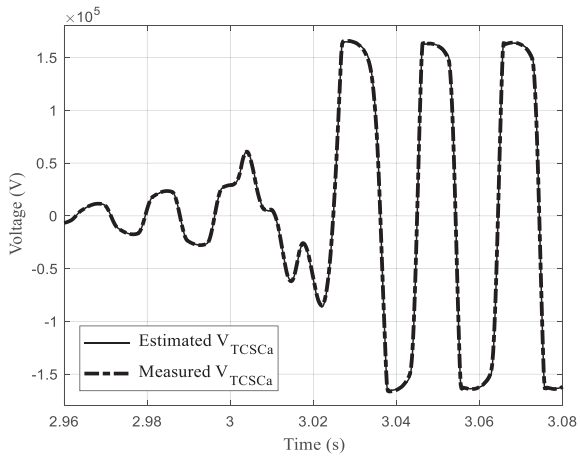


Fig. 26. Measured and estimated voltage drops though TCSC&MOV with 49° firing angle in phase A and C-Syn-C.

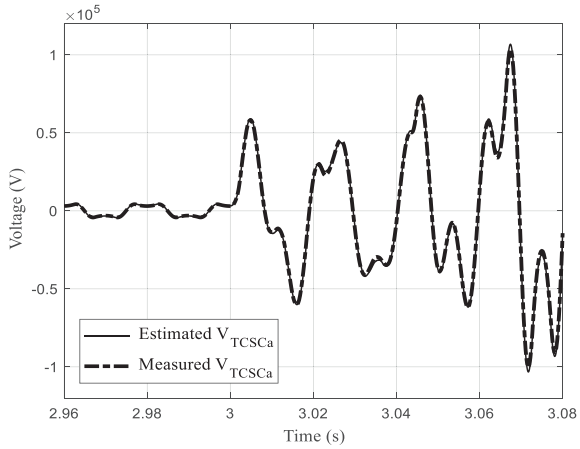


Fig. 27. Measured and estimated voltage drops though TCSC&MOV with 45° firing angle in phase A and I-Syn-V.

range compared with that presented in [15].

5.6. Effect of load angle, noise and sampling frequency changes on the proposed algorithm performance

In this section, the developed algorithm is tested under different operating conditions related to the operation with higher power angle δ, under different noise levels and with different sampling frequencies.

The algorithm is used to estimate the voltage across the TCSC in I-Syn-V when the load angle equals 20°. The three-phase current signals at relay point, at bus A, are shown in Fig. 28 for a solid three-phase fault under these conditions. Figs. 29 and 30 illustrate that the measured and estimated currents of MOVs, series capacitor and TCR are very close to each other. Fig. 30 illustrates the TCR current, where after the fault instant, the inductor current is approximately continuous for three periods and, then, the conduction angle coincides with the applied pulses as shown in Fig. 31. Fig. 32 illustrates the measured and estimated TCSC voltages with the following statistical values: the percentages of maximum, minimum, mean, and standard deviation errors of the estimated voltage drop at 40° firing angle and $\delta = 20°$ are 2.691%, 0.0027%, 1.253%, and 0.916%, respectively. It can be noticed that the MOVs current approximately equals zero, due to the inductive mode and the synchronization with voltage. Therefore, the TCSC voltage during fault is less than 150 kV as shown in Fig. 27. According to Table 2, the error in the estimated voltage on TCSC doesn't exceed 2.69%, while the

Table 1
Summary of the maximum, minimum, mean, and standard deviation errors of the estimated voltage drop for different cases over one cycle.

Mode	α (deg)	Max error %	Min error%	Mean error %	Std. error %
C-Syn-C	70	1.5054	8.83×10^{-4}	0.8897	0.4374
	80	1.5054	8.84×10^{-4}	0.8897	0.4374
	90	2.2817	4.42×10^{-4}	0.8786	0.8571
C-Syn-V	70	1.5070	9.11×10^{-4}	0.7721	0.4549
	80	1.5070	9.11×10^{-4}	0.7721	0.4549
	90	1.5054	8.85×10^{-4}	0.8897	0.4374
I-Syn-C	40	2.6	5.76×10^{-4}	2.0884	0.7439
	20	0.8389	8.82×10^{-4}	0.5827	0.2666
	0	0.4256	6.56×10^{-4}	0.3158	0.1154
I-Syn-V	40	1.9037	3.5×10^{-3}	1.3420	0.4097
	20	0.8215	7.69×10^{-4}	0.5653	0.2612
	0	0.6523	1.77×10^{-4}	0.2765	0.1997
Resonance C-Syn-C	49	1.5060	12×10^{-4}	0.8407	0.4250
Resonance C-Syn-V	49	1.5318	3×10^{-3}	0.6305	0.4960
Resonance I-Syn-C	45	1.7408	4.25×10^{-4}	1.3294	0.4305
Resonance I-Syn-V	45	1.5496	6.8×10^{-3}	0.7814	0.4279

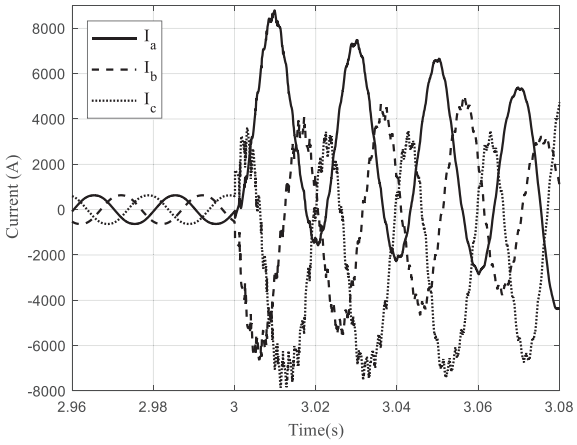


Fig. 28. The waveform of the currents at Bus A under solid 3-phase fault at 180 km and TCSC with 40° firing angle and I-Syn-V at load angle $\delta = 20°$.

error in the fault location equals only 0.57% (where, $Fault\ location\ \ error = \frac{Actual\ \ value - Measured\ \ value}{Transmission\ \ line\ \ length}$).

The algorithm is used to estimate the voltage across the TCSC in I-Syn-V with the load angle of 20°, while the measured signal is applied by 40 dB noise using a white Gaussian noise (WGN) [31]. The three-phase current signals at relay point, at bus A, are shown in Fig. 33 for solid three-phase faults in these conditions. Figs. 34 and 35 illustrate the measured and estimated currents of MOVs, series capacitor and TCR. The estimated current by the developed algorithm is close to the average of the clear measured value. This means that the developed algorithm has the proper ability to filter the noise attack. Fig. 35 illustrates the TCR current, where after the fault instant, the inductor current is approximately continuous for three periods and, then, the conduction angle

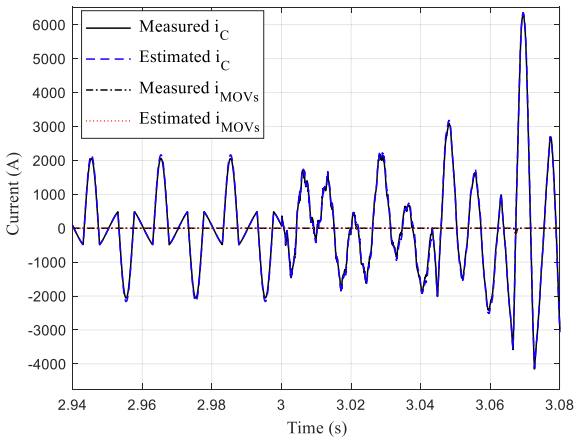


Fig. 29. Capacitor and MOVs current waveforms under solid 3-phase fault at 180 km and TCSC with 40° firing angle and I-Syn-V at load angle $\delta = 20°$.

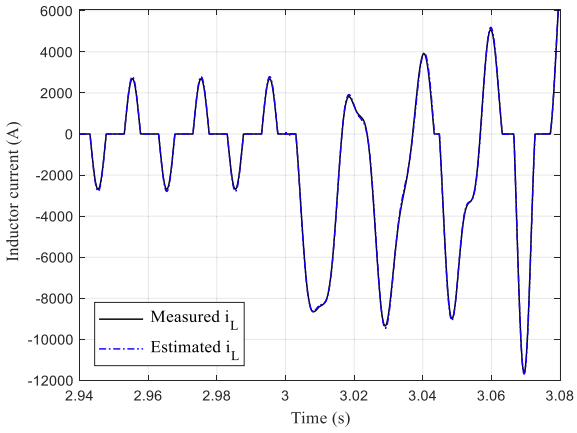


Fig. 30. Inductor current waveform under solid 3-phase fault at 180 km and TCSC with 40° firing angle and I-Syn- V at load angle $\delta = 20°$.

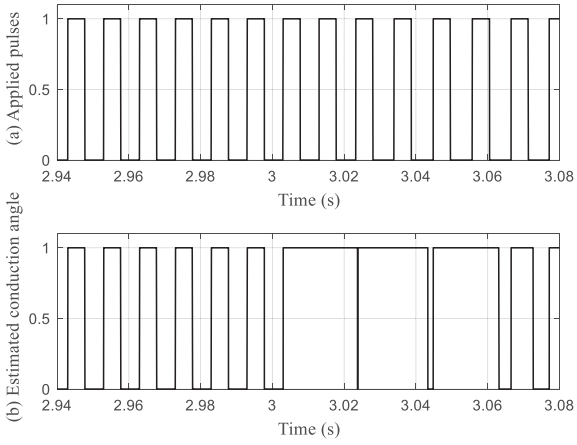


Fig. 31. Applied pulses and actual conduction angles in phase A under three-phase fault at 180 km and TCSC with 40° firing angle and I-Syn-V at load angle $\delta = 20°$.

coincides with the applied pulses as shown in Fig. 36. Although the error in the estimated current affects the estimation of the conduction angle, the application of the developed algorithm overcomes this error. Fig. 37 illustrates the measured and estimated TCSC voltages with the following statistical values: the percentages of maximum, minimum, mean, and

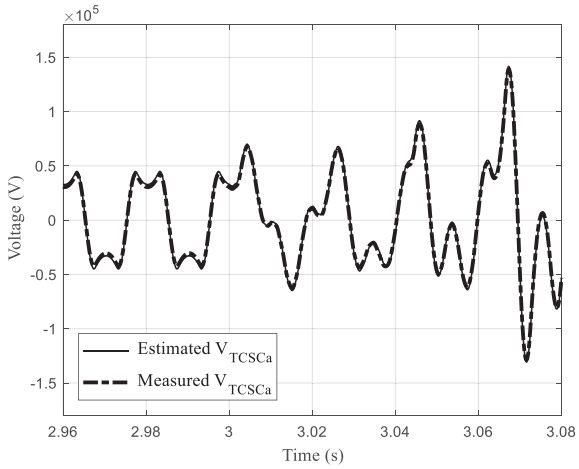


Fig. 32. Measured and estimated voltage drops though TCSC&MOV with 40° firing angle in phase A and I-Syn-V at load angle $\delta = 20°$.

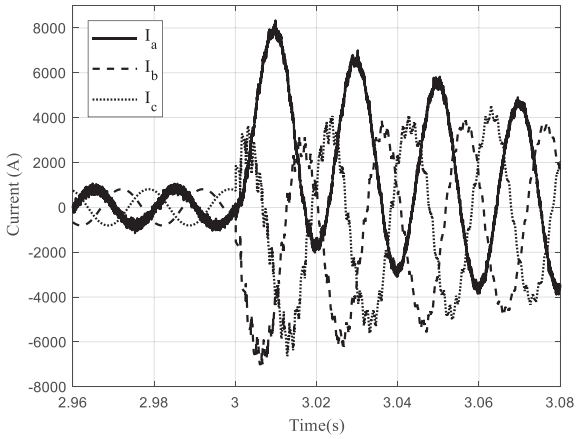


Fig. 33. The waveform of the currents at Bus A under solid 3-phase fault at 180 km and TCSC with 40° firing angle and I-Syn-V at load angle $\delta = 20°$ and 40 dB noise.

standard deviation errors of the estimated voltage drop at 40° firing angle, δ = 20° and 40 dB noise are 2.952%, 0.0045%, 1.378%, and 0.845%, respectively. Table 2 illustrates the statistical evaluation of the proposed algorithm at different noise values of 20, 40 and 60 dB. As shown in Table 2, the error in the estimated voltage on TCSC due to noise does not exceed 4.037%, while the error in the fault location is 0.646%.

The sampling frequency of the measured signal to estimate the required values plays an important role in the accuracy and speed of decision making [11,32]. In the previous studied cases, a sampling frequency of 20 kHz is used. This rate of sampling is considered little bit high, but it provides proper accurate estimation. To illustrate the effectiveness of the proposed technique on various sampling rate, a sampling frequency of only 2 kHz is used. Fig. 38 illustrates the measured and estimated TCSC voltages at 2 kHz sampling rate with the following statistical values: the percentages of maximum, minimum, mean, and standard deviation errors are 3.8408%, 0.026%, 1.6% and 0.882%, respectively. Table 2 illustrates the evaluation of the proposed technique under different sampling frequencies of 10, 4 and 2 kHz. It can be noticed that the decreasing of the sampling frequency leads to increasing of errors in the estimated TCSC voltage and the estimated fault location.

5.7. Algorithm validation for un-transposed line

In this section, the developed algorithm is used to estimate the voltage across the TCSC for un-transposed transmission line. The installation of un-transposed transmission line is not recommended, especially in these long lines, which transmit high power transfer ratings. The un-transposed transmission lines suffer from unbalanced operation. In this study, the suggested algorithm for TCSC voltage estimation is tested when it is installed in un-transposed transmission line to show its effectiveness. Appendix A provides the parameters of the un-transposed transmission line of impedance and admittance matrices per km, i.e., Z_{abc} and Y_{abc}, respectively. The three-phase current signals are shown in Fig. 39 for solidly three phase fault at 200 km, and the TCSC with 80° firing angle and the C-Syn-V at load angle $\delta = 20°$. Fig. 40 illustrates the measured and estimated three phase voltages of TCSC. As shown in this figure, the three phase currents are different for the healthy and faulty cases due to the asymmetrical configuration. The estimated voltages by the developed algorithm are close to the measured values, whether in the healthy or faulty case. The percentages of maximum, minimum, mean, and standard deviation errors of the

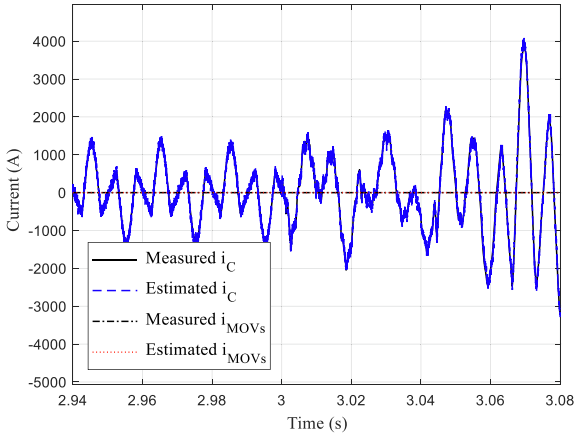


Fig. 34. Capacitor and MOVs current waveforms under solid 3-phase fault at 180 km and TCSC with 40° firing angle and I-Syn-V at load angle $\delta = 20°$ and 40 dB noise.

Table 2
Effect of different noise levels and sampling frequency on the errors of the estimated voltage drop (I-Syn-V mode) and the distance relay performance under solid-LLL fault at 180 km, α = 40° and $\delta = 20°$.

Sampling frequency kHz	Noise level dB	Estimated voltage drop errors				Location	
		Max error%	Min error%	Mean error%	Std. error%	Measured (km)	Error %
20	0	2.691	0.0027	1.253	0.916	178.28	0.572
20	20	2.628	0.0012	1.249	0.806	178.26	0.646
20	40	2.952	0.0045	1.378	0.845	178.72	0.425
20	60	4.037	0.0006	1.473	0.908	179.31	0.227
10	0	2.075	0.0036	0.989	0.532	178.76	0.411
4	0	2.357	0.008	1.228	0.68	178.72	0.427
2	0	3.841	0.026	1.6	0.882	178.29	0.568

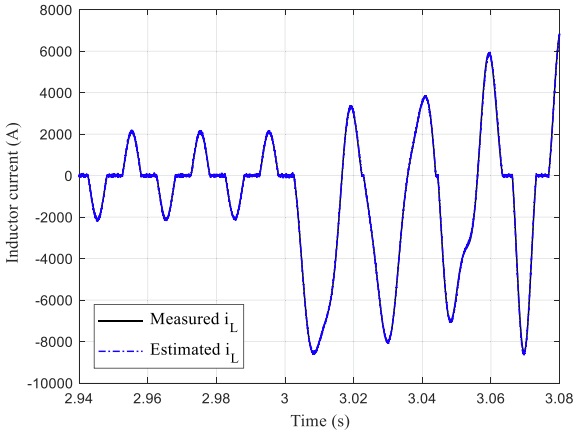


Fig. 35. Inductor current waveform under solid 3-phase fault at 180 km and TCSC with 40° firing angle and I-Syn- V at load angle $\delta = 20°$- and 40-dB noise.

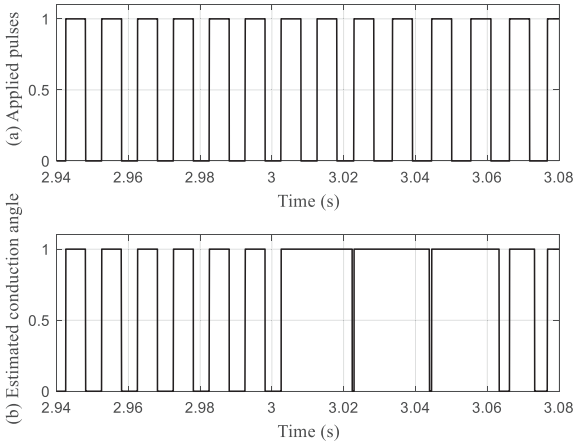


Fig. 36. Applied pulses and actual conduction angle in phase A under three-phase fault at 180 km and TCSC with 40° firing angle and I-Syn-V at load angle $\delta = 20°$ and 40 dB noise.

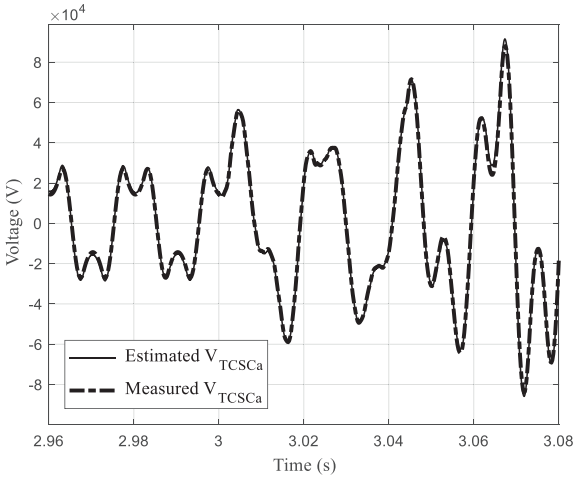


Fig. 37. Measured and estimated voltage drops though TCSC&MOV with 40° firing angle in phase A and I-Syn-V at load angle $\delta = 20°$ and 40 dB noise.

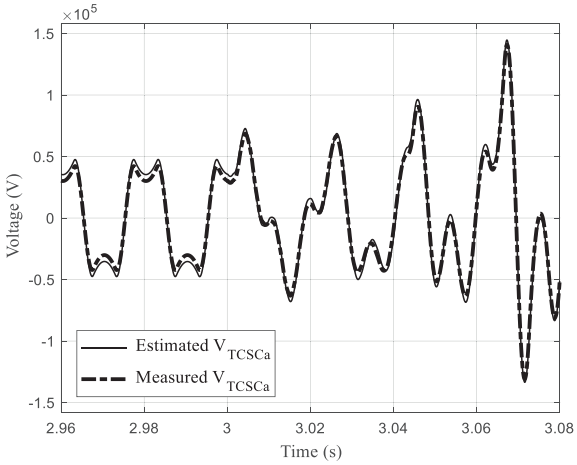


Fig. 38. Measured and estimated voltage drops though TCSC&MOV with 40° firing angle in phase A and I-Syn-V and sampling frequncy 2 kHz.

estimated voltage drop at an 80° firing angle and $\delta = 20°$ are 1.46%, 9.3 × 10–4%, 0.4603%, and 0.4386%, respectively, for all measured phases.

5.8. *Performance of the adaptive distance relay integrated with the proposed algorithm under several stress conditions*

In this work, the estimated TCSC voltage by the proposed algorithm is integrated with the distance relay installed at bus A. It is assumed that system B in Fig. 3 consists of an additional transmission line with a length of 150 km with the same parameters of the essential transmission line (300 km). The following study investigates the performance of the developed algorithm integrated with the distance relay under different common challenges that face the distance relay protection. For example, the proposed technique performance is investigated under stress conditions such as infeed condition, over-reach, load encroachment, voltage instability, and power swing.

Fig. 41 shows the apparent impedance (*ZR*) of the relay under the effect of the infeed with and without the TCSC voltage estimation for a single line to ground (SLG) fault with 30 Ω fault resistance at 200 km from bus A. As shown in Fig. 41, the distance relay correctly detects the fault in Zone 1 for all no infeed (blue and red lines) cases either the voltage across the TSCS is considered or not. It can be noticed that, for the red line trajectory, neglecting TCSC voltage reduces the accuracy of estimated line impedance because the red line slightly enters zone 1.

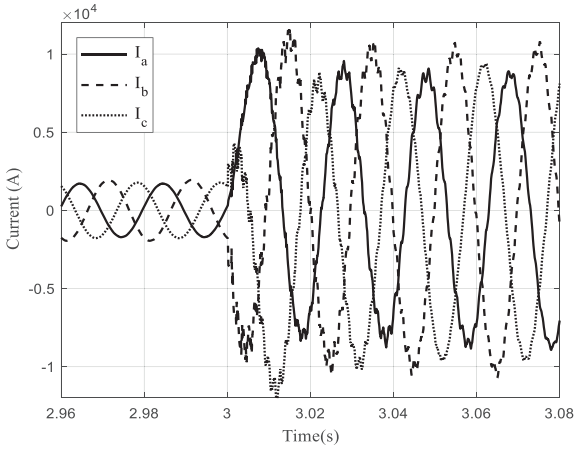


Fig. 39. The waveform of the currents at Bus A under solid 3-phase fault at 200 km and TCSC with 80° firing angle and C-Syn-V at load angle $\delta = 20°$ with un-transposed line.

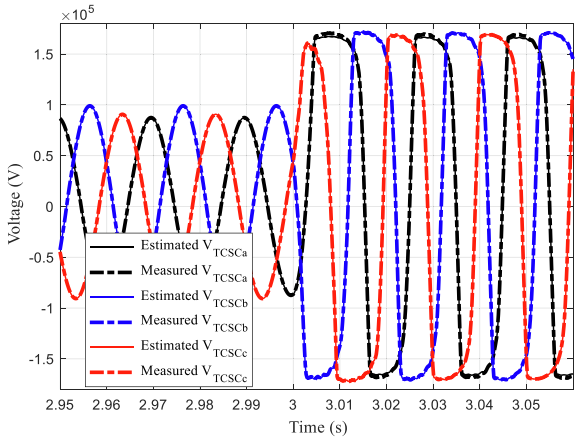


Fig. 40. Measured and estimated voltage drops though TCSC&MOV under solid 3-phase fault at 200 km and TCSC with 80° firing angle and C-Syn-V at load angle $\delta = 20°$ with un-transposed line.

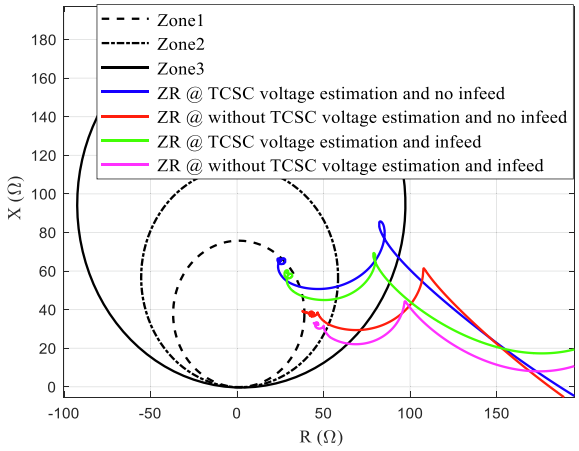


Fig. 41. Apparent impedance of the relay for the effect of infeed with and without TCSC voltage estimation at SLG fault with 30 Ω fault resistance at 200 km.

Table 3
The fault resistance effect on the error of the estimated voltage drop (I-Syn-V mode) and the distance relay performance under different fault types at 200 km.

Fault type	R_f (Ω)	Estimated voltage drop errors %	Location	
			Measured (km)	Error %
LLL	0	0.7333	199.11	0.297
	10	0.6687	187.31	4.23
	30	0.4383	203.37	1.125
	50	0.2520	215.1	5.033
L-G	0	0.3233	199.97	0.009
	10	0.3099	195.02	1.661
	30	0.2188	201.99	0.666
	50	0.1224	219.62	6.54
LL	0	0.423	199.11	0.297
	10	0.267	189.18	3.6
	30	0.238	205.4	1.8
	50	0.225	217.25	5.75
LLG	0	0.5233	199.01	0.33
	10	0.432	191.11	2.96
	30	0.315	197.96	0.68
	50	0.195	213.5	4.5

However, for infeed cases revealed in Fig. 41, the behavior is changed, where, the distance relay correctly detects the fault in Zone 1 when the voltage across the TSCS is considered (green line). The distance relay is in under-reach case and detects the fault in Zone 2 when the voltage across the TSCS is not considered (purple line). Table 3 shows the fault resistance effect on the error of the estimated voltage drop (I-Syn-V mode) and the distance relay performance under different fault types at 200 km for infeed case. As mentioned in Section 4.3, the fault location is obtained via dividing the imaginary part of calculated impedance per line reactance per unit length, to reduce the fault location error. As shown in Table 3, the fault location error is increased by increasing the fault resistance for different fault types. In all cases, the TCSC voltage estimation has a low value of error. However, and in general, the voltage estimation reduces the error of distance estimation as shown in Fig. 41. The maximum fault location error is 6.54% at 50 Ω fault resistance, which is considered as an acceptable range. However, the estimated voltage drop error is decreased with increasing the fault resistance and it is almost constant for different fault types. It is concluded that the fault location is not essentially affected by the estimation of the voltage across the TCSC.

Fig. 42 shows the relay apparent impedance under a solid three phase fault at 270 km (an external fault with respect to Zone 1) with and without the TCSC voltage estimation. In addition, TCSC is operated at different modes as I-Syn-V with 20° firing angle and C-Syn-V with 80° firing angle. As shown in Fig. 42, the relay correctly detects the faults at Zone 2 for I-Syn-V whether the voltage across the TSCS is considered or not. On the other hand, the trajectory of the relay enters Zone 1 for about 0.002 s and then settles down correctly at zone 2 for C-Syn-V when the voltage across the TSCS is considered. However, the trajectory of the relay impedance is finally located in zone 1, incorrect over-reach case, for C-Syn-V when the voltage across the TSCS is neglected.

The load encroachment case is investigated by increasing the active power demand of the load at the receiving end [32]. To evaluate the performance of the distance relay with the proposed integrated algorithm, the load is suddenly increased up to 200% of its initial value as a step change at 2.95 s. Then, a solid 3-phase fault is applied at 3 s. Fig. 43 shows the measured and estimated three phase TCSC and MOV voltages with 20° firing angle in phase A and I-Syn-V under load encroachment effect. The estimated voltage by the developed algorithm is close to the measured value either in the normal, load change or faulty cases. Fig. 44 shows the relay apparent impedance for the effect of the load encroachment and solidly three-phase fault at 200 km compared with fault case only. In addition, different modes are compared I-Syn-V with 20° firing angle and C-Syn-V with 80° firing angle. As shown, the relay correctly detects the faults at Zone 1 for all modes whether the voltage across the TSCS is considered or not.

Voltage instability case is developed by increasing the reactive

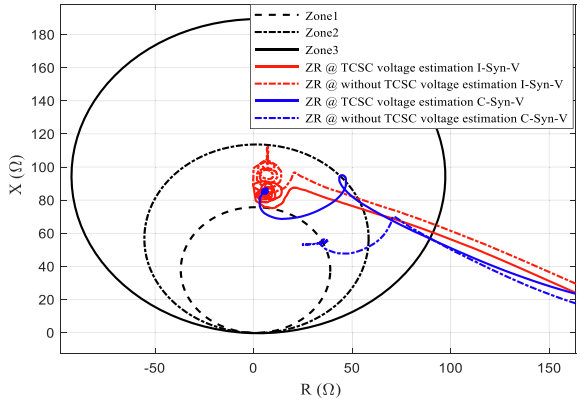


Fig. 42. Apparent impedance of the relay under solid 3-phase fault at 270 km (an external fault with respect to zone 1) with and without TCSC voltage estimation.

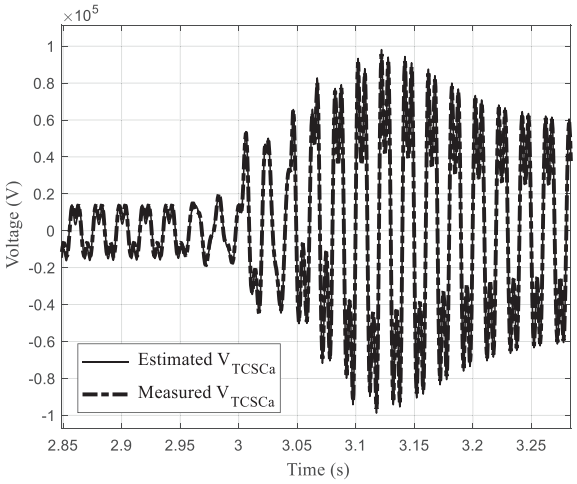


Fig. 43. Measured and estimated voltage drops across TCSC&MOV with 20° firing angle in phase A and I-Syn-V and load encroachment effect.

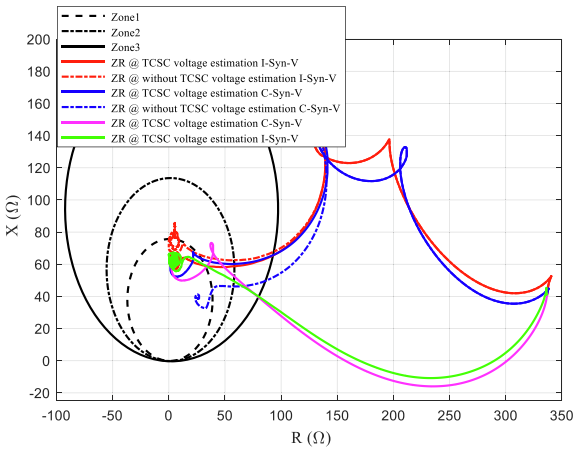


Fig. 44. Apparent impedance of the relay for the effect of load encroachment (load increasing by 200% at 2.95 s and solid 3-phase fault at 3 s at 200 km).

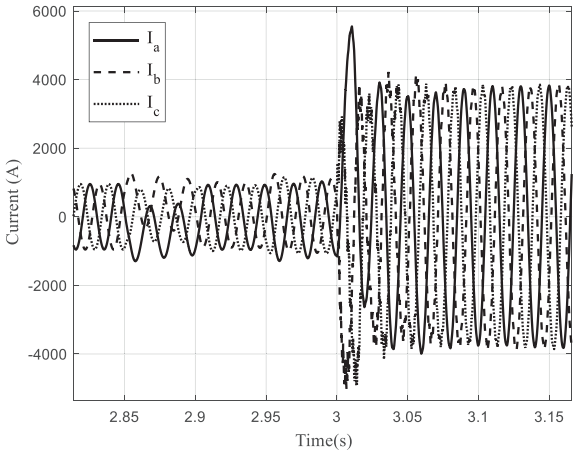


Fig. 45. The waveform of the currents at Bus A for solid 3-phase fault at 200 km and TCSC I-Syn-V with 20° firing angle during voltage instability.

power demand of the load at the receiving end [32]. This is accomplished by increasing the reactive power up to 200, 300 and 400% of its value as a step change at 2.85, 2.9 and 2.95 s, respectively. Consequently, a solidly three phase fault is applied at 3 s. Fig. 45 illustrates the

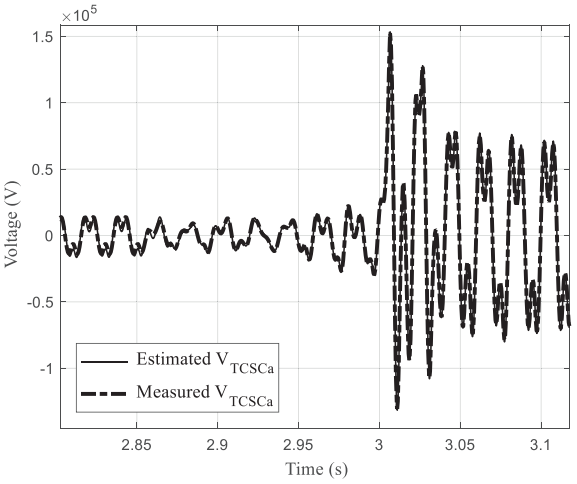


Fig. 46. Measured and estimated voltage drops across TCSC&MOV with 20° firing angle in phase A and I-Syn-V during voltage instability.

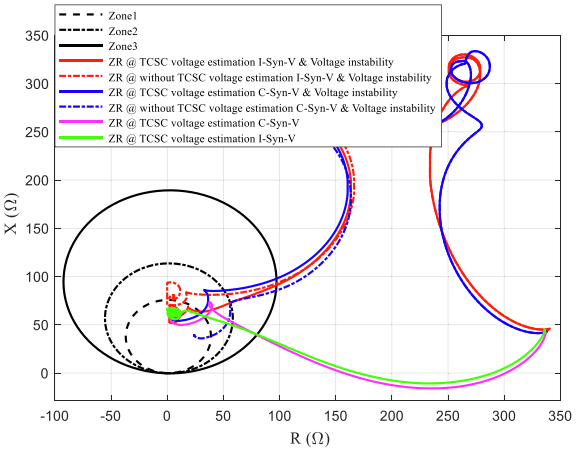


Fig. 47. Apparent impedance of the relay during voltage instability.

waveform of the currents at Bus A for the solidly three-phase fault at 200 km and the TCSC is in I-Syn-V with 20° firing angle during voltage instability. Fig. 46 illustrates the measured and estimated three phase voltages of TCSC and MOV with 20° firing angle in phase A and I-Syn-V and voltage instability effect. Fig. 47 shows the apparent impedance of

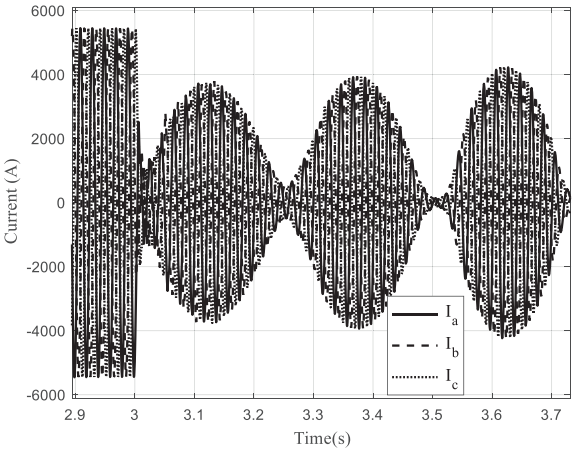


Fig. 48. The waveform of the currents at Bus A for TCSC C-Syn-V with 80° firing angle during unstable power swing.

the relay due to the effect of voltage instability and solidly three-phase fault at 200 km compared with the faulty case without voltage instability. In addition, different modes are illustrated to compare I-Syn-V with 20° firing angle and C-Syn-V with 80° firing angle. As shown in Fig. 47, the relay correctly detects the faults at Zone 1 for all modes whether the voltage across the TSCS is considered or not.

Finally, to evaluate the distance relay behavior during power swing with the TCSC in different operation modes, a solidly three phase fault occurs at 180 km at 2.6 s and is cleared at 3 s (takes 20 cycles) [14,32]. Fig. 48 illustrates the waveform of the currents at Bus A during power swing and TCSC C-Syn-V with 80° firing angle. Fig. 49 shows the measured and estimated TCSC and MOV voltages for this case. The estimated voltage by the proposed algorithm is close to the measured value. Fig. 50 shows the apparent impedance of the relay during unstable power swing for different modes at I-Syn-V with 20° firing angle and C-Syn-V with 80° firing angle. As shown, the relay behaves correctly and does not indicate any false trip action under the swing that occurs due to the swing condition. As shown in Fig. 50, a false trip is indicated at Zone 3 for C-Syn-V mode considering the voltage estimation only. The distance relay behaves correctly as it does not falsely trip for this swing as well as for the inductive mode.

6. Conclusions

In this paper, a precise generalized technique for voltage estimation of TCSC has been developed to adapt the performance of distance relay. The principle of operation of the TCSC has been introduced to illustrate its effect on the operation of transmission system. The TCSC modes of operation and methods of synchronization have been explained to provide the whole spectrum of operation circumstances. The TCSC has been design for 400 kV two-machine model overhead transmission line test system. The numerical discrete model of the TCSC&MOV system has been developed to estimate both the conduction angle and the voltage across them. Based on the estimated voltage, the algorithm succeeded to detect the fault location according to the system parameters. The developed algorithm has proved high capability to estimate the voltage across the TCSC whether it operates in capacitive, inductive or resonance modes and with current or voltage synchronization methods. The maximum error difference between the estimated and actual signal does not exceed 4.037% of all studied cases that include the signals attacked by noise. Moreover, the distance relay integrated with the proposed algorithm has a maximum fault location error that does not exceed 6.54% in all studied cases, i.e., around 500 cases. The proposed technique has been tested under different loading conditions and modes of operation. It has provided accurate and proper operation under different sampling rates and applied noise. The conventional distance relay with three zones has been adapted by the application of the estimated voltage across TCSC to provide accurate operation. The combination between the distance relay and the proposed technique has been tested under different operating conditions such as over-reach, under-reach, load encroachment, voltage instability, and power swing cases. The results have proved the positive effect of TCSC voltage estimation on the distance relay operation to operate correctly in all studied cases. Therefore, the obtained results have proved the accuracy of the generalize developed algorithm to adapt the performance of the distance relay.

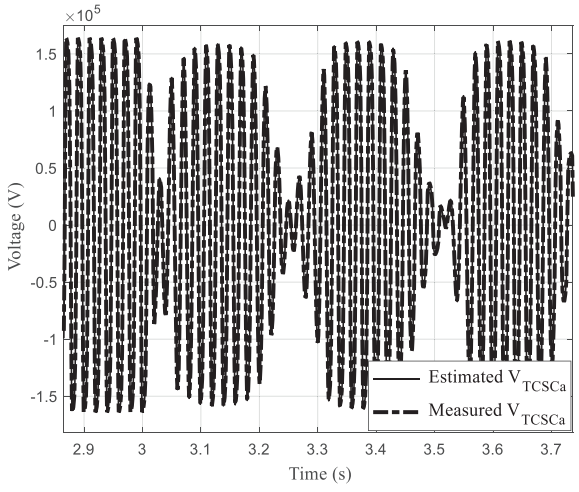


Fig. 49. Measured and estimated voltage drops across TCSC&MOV with 80° firing angle in phase A and C-Syn-V during unstable power swing.

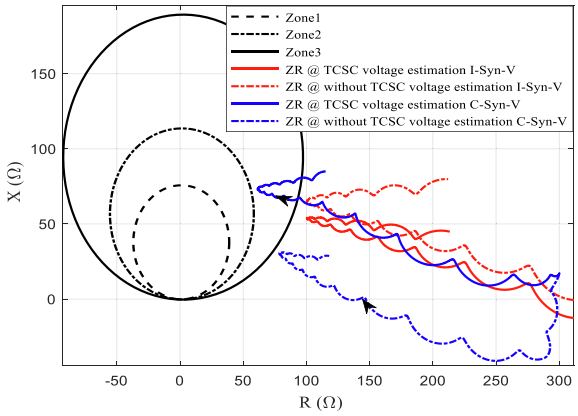


Fig. 50. Apparent impedance of the relay during unstable power swing.

CRediT authorship contribution statement

Ibrahim B.M. Taha: Conceptualization, Methodology, Visualization, Supervision. **Ahmed E. ELGebaly:** Conceptualization, Software, Validation, Formal analysis, Writing - original draft. **Eman S. Ahmed:** Conceptualization, Methodology, Investigation, Formal analysis. **Hossam A. Abd el-Ghany:** Methodology, Software, Validation, Investigation, Resources, Data curation, Writing - review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A

The parameters of the TL are as follows:

- The positive impedance equals (0.0267 + j0.3151) Ω/km.
- The zero-sequence impedance equals (0.275 + j1.0265) Ω/km.
- The total distance is 300 km

The TL connects two power sources that have the following parameters:

- 400 kV rated line to line voltage,
- Nominal frequency $f = 50$ Hz, or $\omega_N = 2\pi f$ of 315 rad/s,
- Positive- and zero-sequence impedances equal (0.656 + j7.5) and (1.16 + j 11.25) Ω, respectively,
- The power angle δ between the two sources is 2°.

The TCSC parameters are as follows:

- $C = 63$ μF
- $L = 25.7$ mH
- $\lambda = 2.5$.

The MOV's current is $P\left(\frac{v_x}{v_{ref}}\right)^q$ [6]. The parameters of MOV are:

- q equals 23,
- $P$is equal to 1 kA,
- The reference voltage (v_{ref}) is equal to 150 kV.

The impedance matrix of the un-transposed transmission line per unit length (Ω/km) is given as follows:

$$Z_{abc} = \begin{bmatrix} 2.67 + 31.416j & 2.43 + 14.98j & 2.37 + 12.28j \\ 2.43 + 14.98j & 2.75 + 31.3j & 2.43 + 14.98j \\ 2.37 + 12.28j & 2.43 + 14.98j & 2.67 + 31.416j \end{bmatrix} *10^{-2}$$

The admittance matrix of the un-transposed transmission line per unit length (Ω/km) is given as follows:

$$Y_{abc} = \begin{bmatrix} 40.84j & -7.45j & -2.04j \\ -7.45j & 42.43j & -7.45j \\ -2.04j & -7.45j & 40.84j \end{bmatrix} *10^{-7}$$

The parameters of the un-transposed transmission line are calculated by assuming that the phases are arranged in flat plain, while phase b is between phases a and c.

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